Q **August 11, 2026**

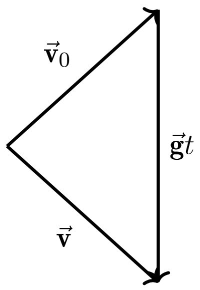
Ryan Huang, Sophie Chen

Vector Triangles and Their Simple Applications **[7 pts]**

The vector triangle is a practical and common tool in physics, yet it barely appears in AP examinations. What follows is a brief introduction, a worked example, and a set of problems for the reader to explore.

Concept and Example

Consider the following scenario. A point mass m is launched from the ground at an angle θ; its initial velocity $\vec{\mathbf{v}}_0$ is the leftmost vector in Figure 1. Throughout the motion a constant downward acceleration of magnitude g acts on the body — that is the vector on the right of the + sign in the figure. At landing the velocity is the vector to the right of the = sign. The whole evolution can therefore be synthesised into the triangle shown on the right.



$$\vec{\mathbf{v}} = \vec{\mathbf{v}}_0 + \vec{\mathbf{g}}t$$

Fig. 1 — The vector triangle for ground-to-ground projectile motion. $\vec{\mathbf{v}}_0$ and $\vec{\mathbf{g}}t$ have fixed directions; $\vec{\mathbf{v}}$ and the flight time t vary with θ.

This construction leads to a classic question: at what angle θ must one launch the mass to maximise the horizontal range? Intuition (and elementary kinematics plus trigonometry) gives 45°. Here we offer a different approach — using the *area* of the vector triangle.

Integrating velocity gives displacement, so let us examine the area of the triangle formed by $\vec{\mathbf{v}}_0$, $\vec{\mathbf{g}}t$, and $\vec{\mathbf{v}}$. A short calculation (left to the reader in Q1 below) shows that

$$S = \tfrac{1}{2}\, gx, \tag{3}$$

where x is the horizontal range. Notice that the area depends *only* on x; once we express S in terms of the launch parameters, maximising x reduces to maximising the area of a triangle with two sides of fixed length.

Indeed, in our triangle one side ($\vec{\mathbf{g}}t$) changes with the flight time, but another view fixes $\vec{\mathbf{v}}_0$ and varies the angle α between $\vec{\mathbf{v}}_0$ and $\vec{\mathbf{v}}$. A second calculation (Q2) yields

$$S = \tfrac{1}{2}\, v_0^2 \sin 2\theta = -\tfrac{1}{2}\, v_0^2 \sin 2\alpha. \tag{4}$$

From either expression the maximum occurs at $\theta = 45°$, recovering the familiar result through a purely geometric route.

Problems

The following exercises walk you through the derivation sketched above and extend the idea to a slightly more general setting.

> **Assumptions**
>
> 1. Flat, level ground; uniform gravity $\vec{\mathbf{g}} = -g\,\hat{\mathbf{y}}$; no air resistance.
> 2. The vector triangle is formed by $\vec{\mathbf{v}}_0$, $\vec{\mathbf{g}}t$, and $\vec{\mathbf{v}}$ at the instant of landing ($y = 0$).
> 3. In Q3 the launch point is elevated: $y(0) = h$, $y(t_{\text{land}}) = 0$.

Questions.

Q1. Area in terms of range. Consider the same scenario as the example above (ground-to-ground). Prove that the area of the vector triangle is

$$S = \tfrac{1}{2}\,gx,$$

where x is the horizontal range of the projectile.

Q2. Area in angular form. For the same scenario, show that the area can also be written as

$$S = \tfrac{1}{2}\,v_0^2 \sin 2\theta = -\tfrac{1}{2}\,v_0^2 \sin 2\alpha,$$

where α is the angle between $\vec{\mathbf{v}}_0$ and $\vec{\mathbf{v}}$. Use either expression to confirm that the range is maximised at $\theta = 45°$.

Q3. Non-zero launch height. Now the mass is launched from $y = h > 0$ and lands at $y = 0$ (all else unchanged). Determine the angle θ that maximises the horizontal distance L. No resistance of any kind.

Hint: the vector-triangle idea generalizes — think about what changes when the launch and landing heights differ.